

# Polling vs Webhooks vs WebSockets

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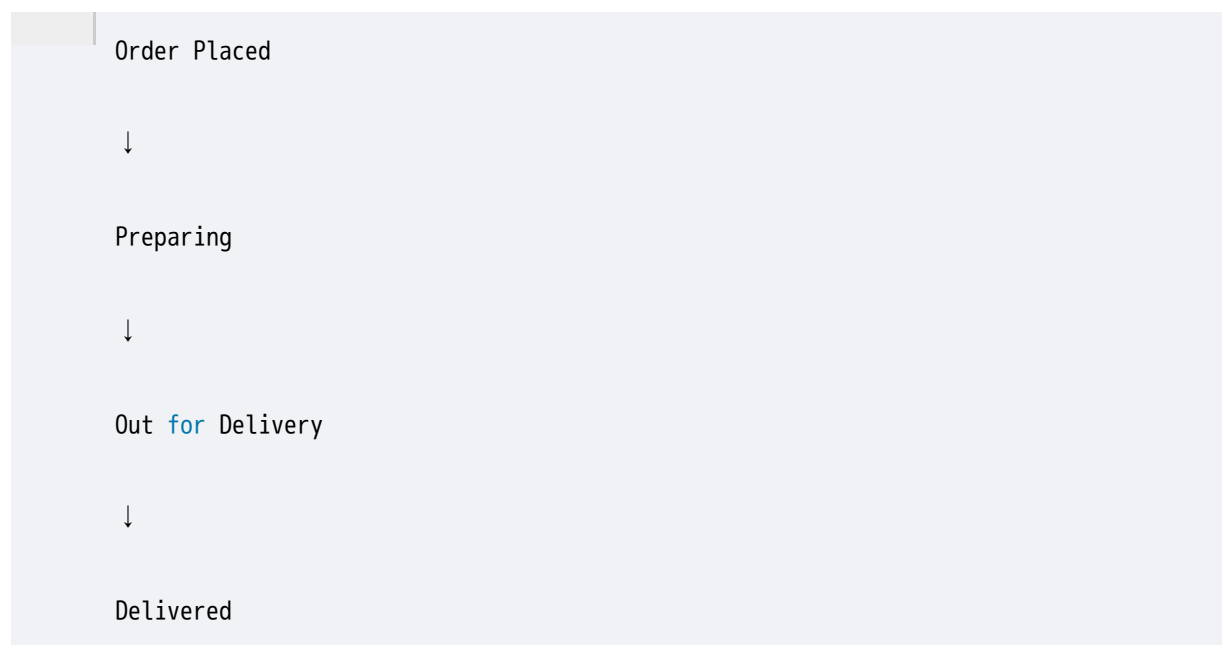
## Chapter 1 — Polling (Zero → Hero)

### 1. The Real Problem

Let's forget technology for a minute.

Suppose you ordered a pizza.

Current status:



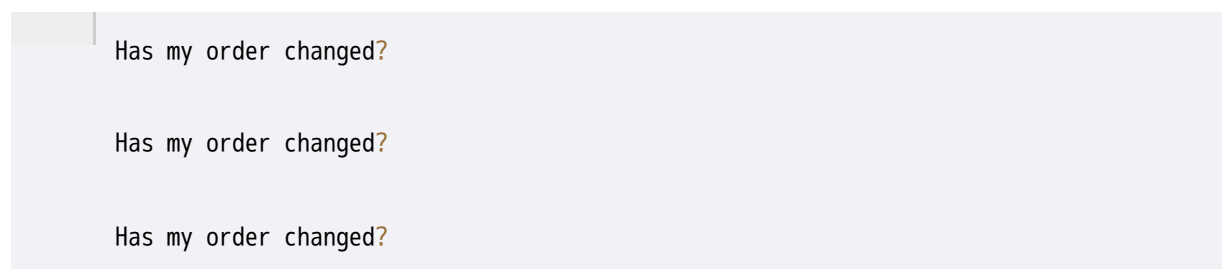
Question:

**How will your mobile app know the status changed?**

There are only two possibilities.

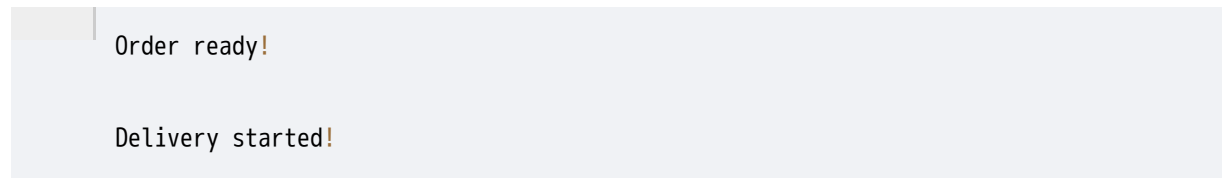
Option 1

The app keeps asking.



## Option 2

Restaurant informs you automatically.



Polling is Option 1.

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## 2. What is Polling?

### DEFINITION

Polling is a communication technique where **the client repeatedly sends requests to the server to check if new data is available.**

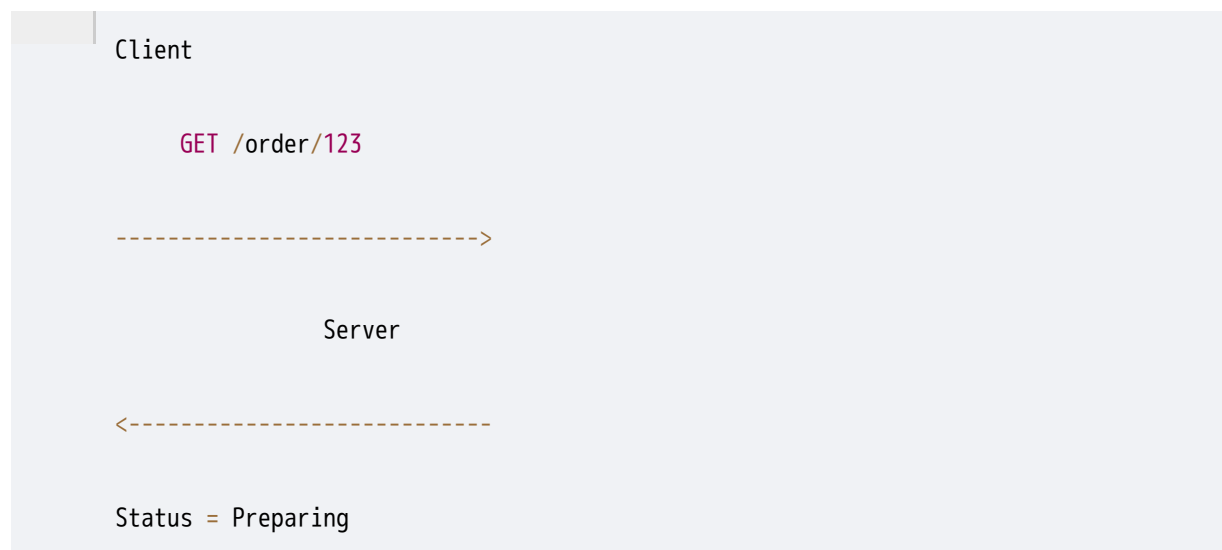
Notice something important.

The server never starts communication.

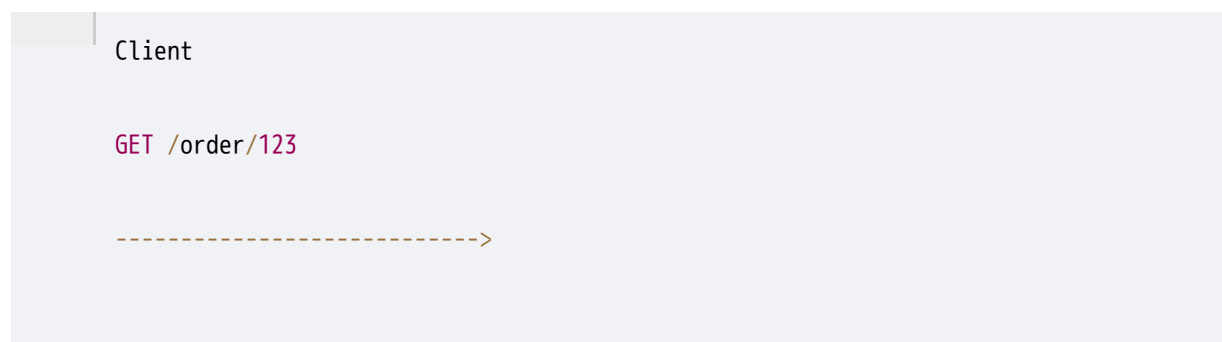
The client always starts it.

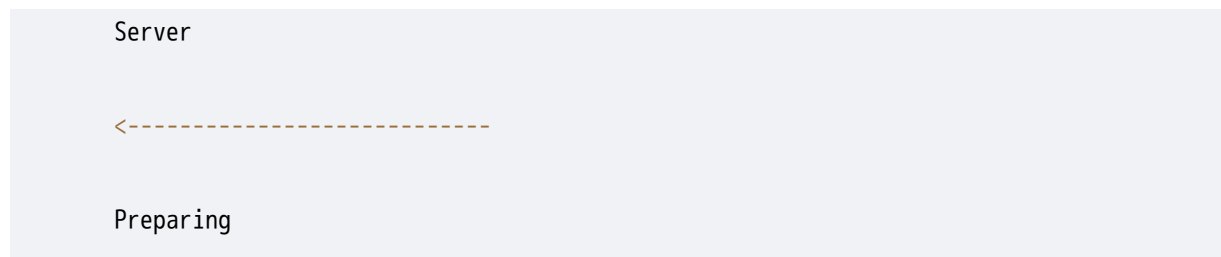
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### Visualization

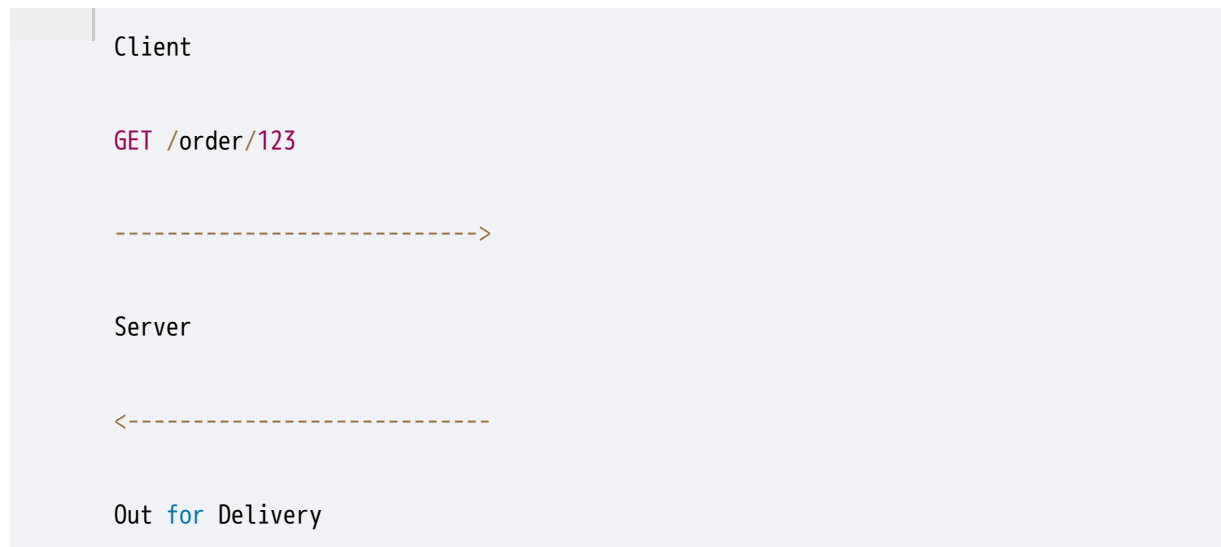


After 5 seconds





Again



The server never pushed anything.

The client kept asking.

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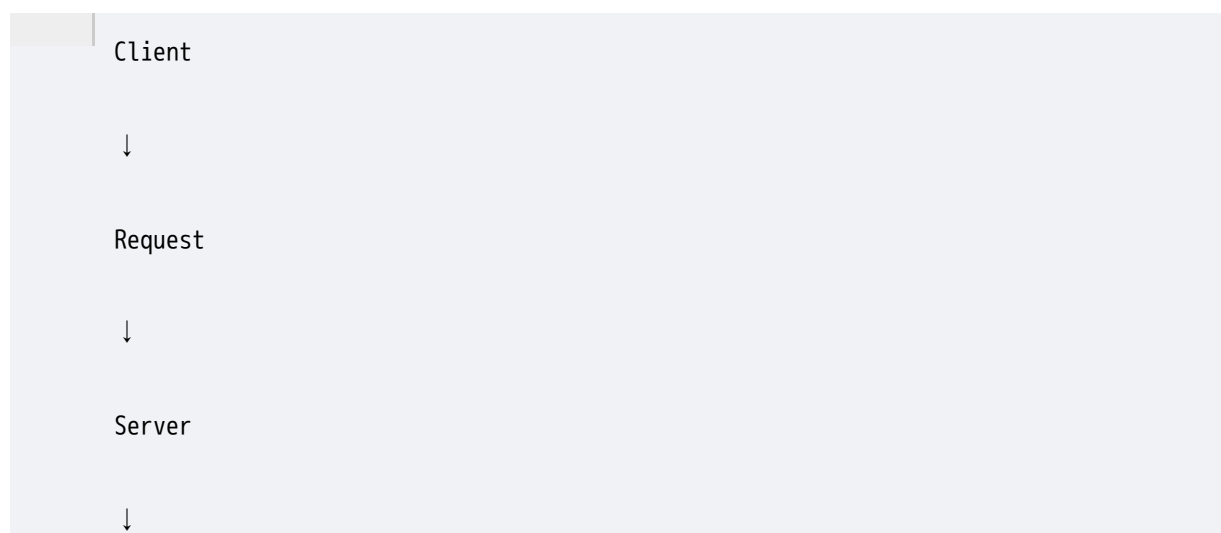
### 3. Why Polling Exists

HTTP is based on the Request → Response model.

The server **cannot randomly send data** to your browser or mobile app.

It can only respond after receiving a request.

HTTP works like this.



Response



Connection Closed

Once the response is sent,  
the server forgets about that request.

This is one of the biggest reasons Polling became popular.

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## 4. Real HTTP Flow

Suppose Zomato app.

```
GET /api/orders/123
```

Server

```
{  
  "status": "Preparing"  
}
```

Five seconds later

```
GET /api/orders/123
```

Server

```
{  
  "status": "Preparing"  
}
```

Five seconds later

```
GET /api/orders/123
```

Server

```
{  
  "status": "Out for Delivery"  
}
```

Notice.

The response is exactly the same twice.

Those requests were unnecessary.

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## 5. Client-side Code

JavaScript example

```
setInterval(async () => {  
  
    const response = await fetch("/api/orders/123");  
  
    const order = await response.json();  
  
    updateUI(order);  
  
}, 5000);
```

Every 5 seconds

↓

HTTP Request

↓

Receive Response

↓

Update UI

Simple.

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## 6. Advantages

**EASY TO IMPLEMENT**

No special protocol.

Every programming language supports HTTP.

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**WORKS EVERYWHERE**

No firewall issues.

No proxy issues.

No browser compatibility issues.

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## STATELESS

Each request is independent.

Server doesn't need to remember previous requests.

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## EASY TO DEBUG

Open Developer Tools.

Network Tab.

You can literally see every request.

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# 7. Disadvantages

This is where interviews become interesting.

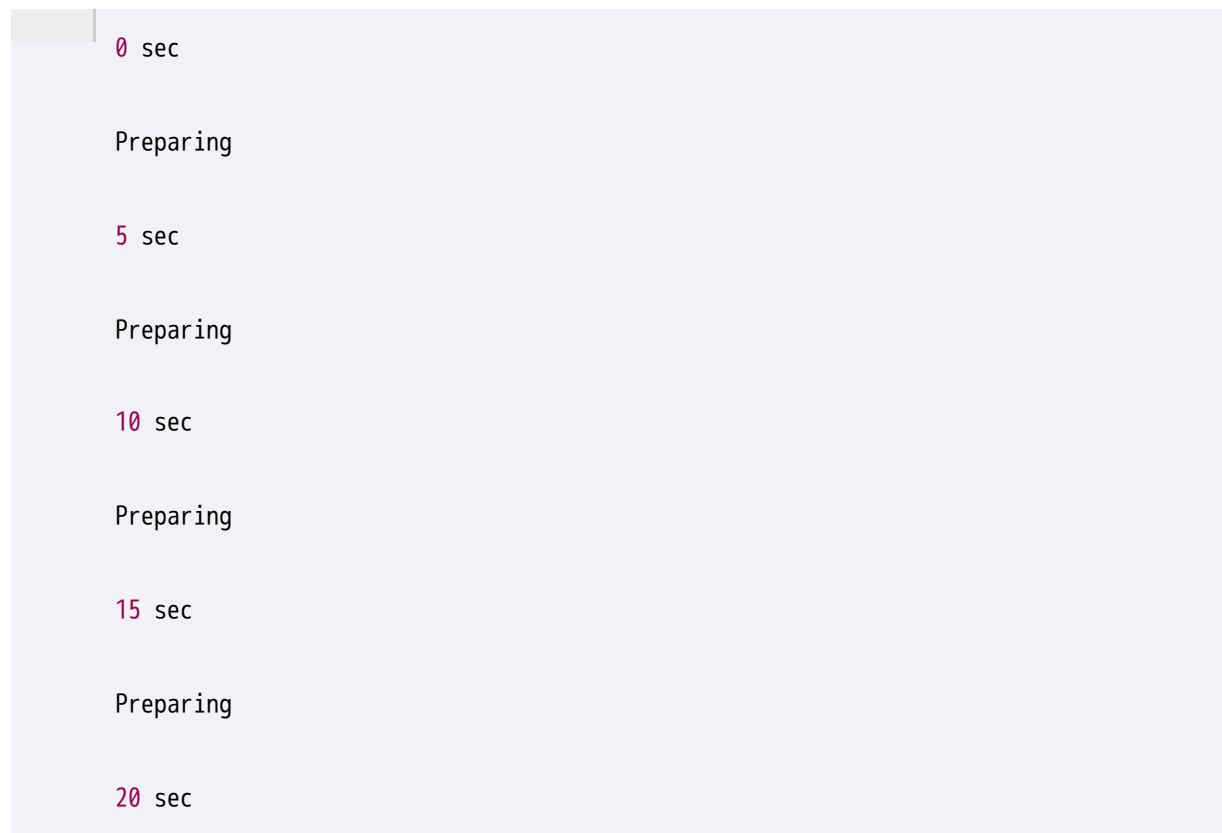
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## PROBLEM 1

Unnecessary Requests

Suppose order status changes every 10 minutes.

But the client polls every 5 seconds.



Almost every request returned the same response.

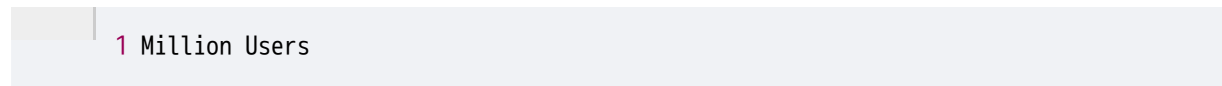
Huge waste.

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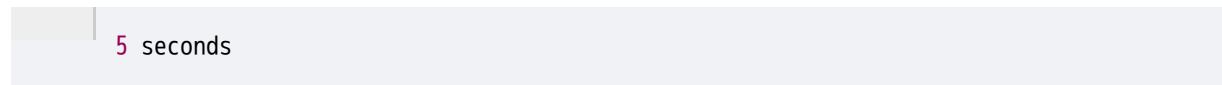
## PROBLEM 2

### Server Load

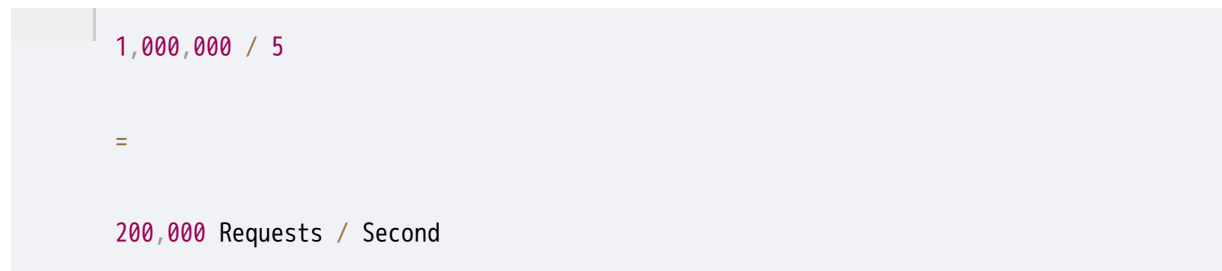
Imagine



Each user polls every



Requests per second



Now imagine

Nothing changed.

Still

200,000 requests every second.

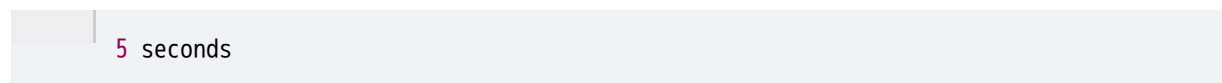
This is called **wasted work**.

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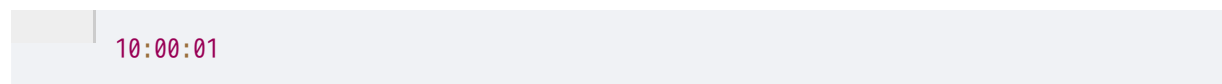
## PROBLEM 3

### Higher Latency

Suppose polling interval



Status changes



Next request comes



User sees update

4 seconds late

Average delay

Polling Interval / 2

This is a common interview point.

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## 8. Short Polling vs Long Polling

Many engineers think Polling means only one thing.

Actually there are two types.

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### A. SHORT POLLING

Most common.

Client

↓

Request

↓

Response

↓

Wait 5 sec

↓

Request Again

This is what most dashboards use.

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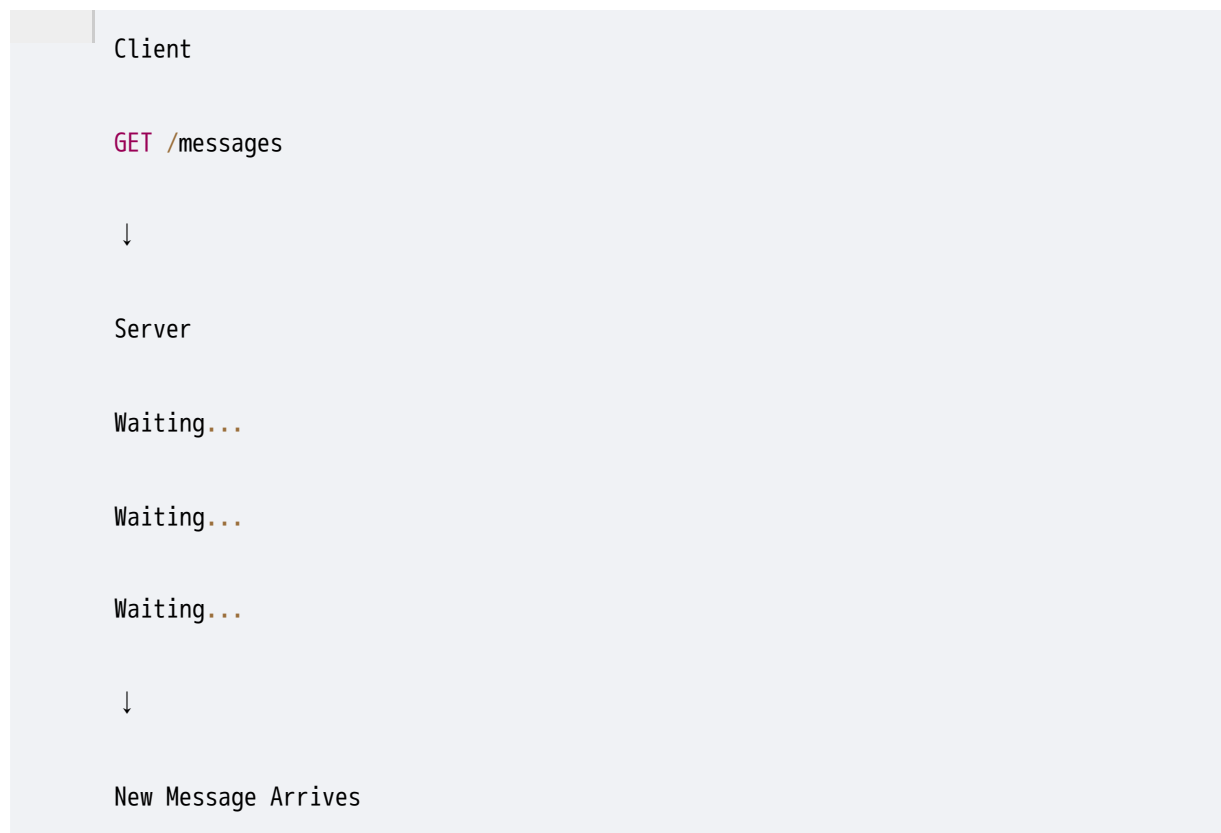
### B. LONG POLLING

Very different.

Flow



## Example



## Immediately

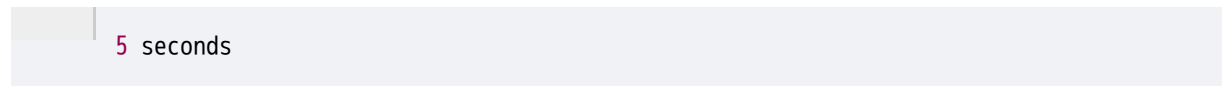
Client sends another request.

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## WHY LONG POLLING?

Because

Instead of checking every



the server waits until

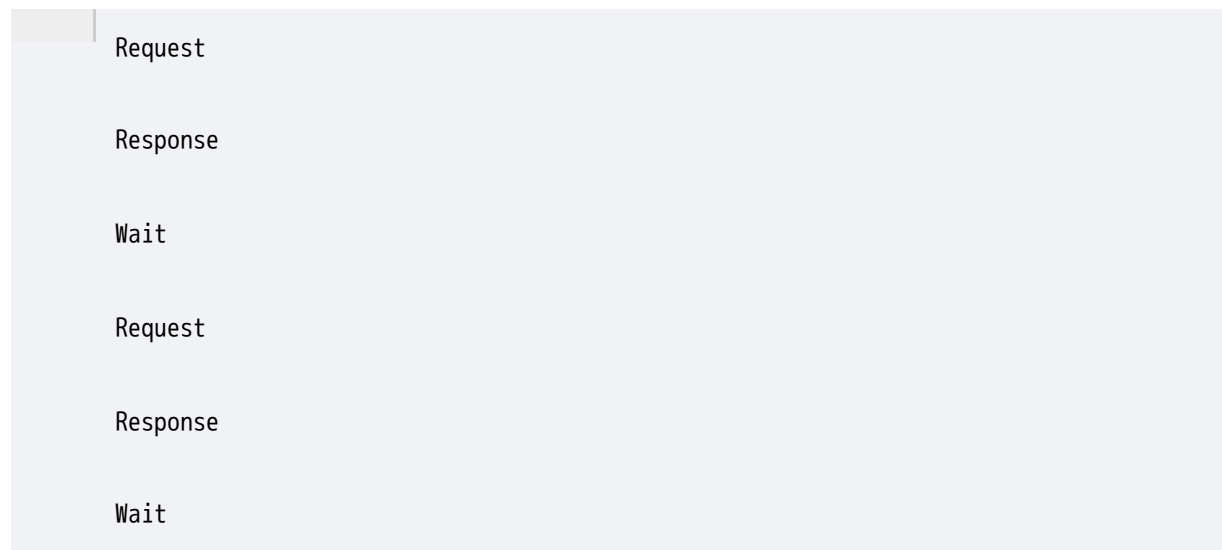
something actually changes.

This reduces unnecessary requests.

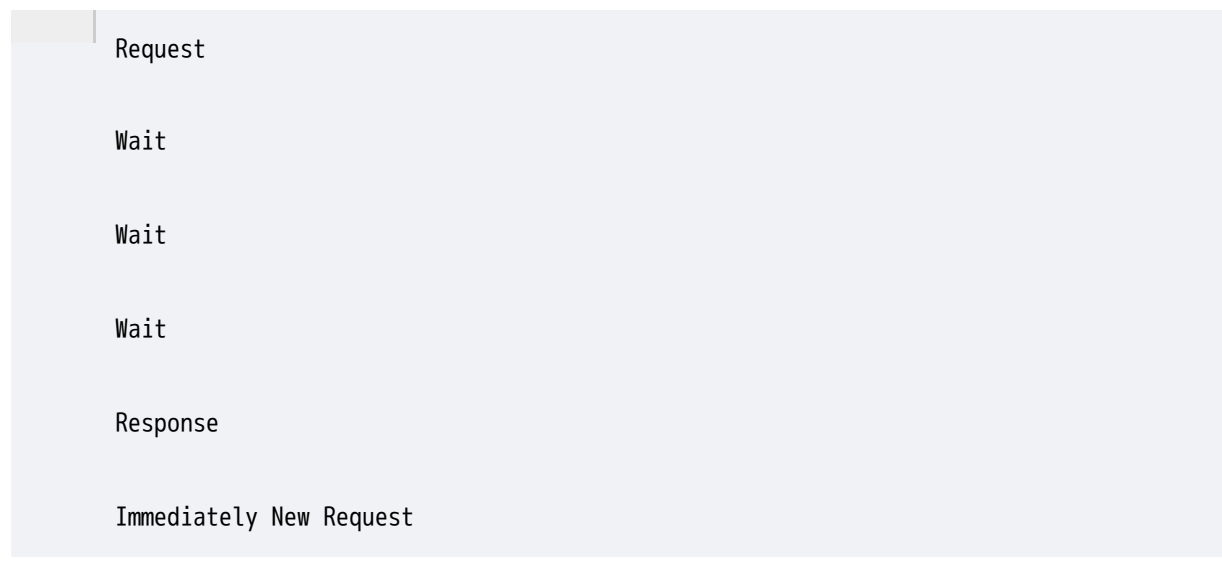
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## Comparison

### Short Polling



### Long Polling



## 9. Where is Polling Used?

Polling is still used today.

People think it is obsolete.

Wrong.

Examples

✓ CI/CD Build Status

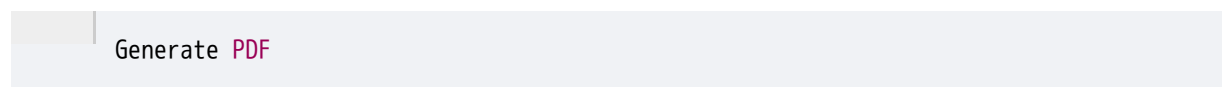
GitHub Actions

Jenkins

Azure DevOps

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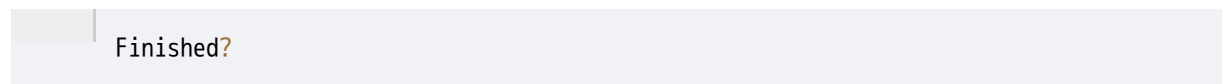
✓ Report Generation



Job runs

2 minutes

Frontend checks



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✓ Data Export

Large Excel

CSV

Background Job

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✓ Admin Dashboards

Internal dashboards

Monitoring tools

Analytics pages

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✓ IoT Devices

Some embedded devices periodically send data to a server instead of maintaining a permanent connection.

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## 10. When NOT to Use Polling

Polling is a poor choice for:

- ✗ WhatsApp Chat
- ✗ Google Docs Collaboration
- ✗ Live Stock Prices
- ✗ Multiplayer Games
- ✗ Video Calls

Reason

Too many unnecessary HTTP requests.

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## 11. Polling Interview Questions

### Q1. WHAT IS POLLING?

Best Answer

Polling is a client-driven communication mechanism where the client periodically sends HTTP requests to check whether new data is available on the server.

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### Q2. WHY IS POLLING INEFFICIENT?

Expected answer

Because

- unnecessary requests
  - higher latency
  - increased server load
  - increased bandwidth usage
- 

### Q3. CAN POLLING BE REAL-TIME?

Technically

Yes.

If polling interval is

100 ms

But

Server will probably hate you 😊

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### Q4. DIFFERENCE BETWEEN POLLING AND LONG POLLING?

Polling

↓

Server responds immediately.

Long Polling

↓

Server keeps request open until data becomes available.

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#### Q5. WHY DO COMPANIES STILL USE POLLING?

Because

- Simple
  - Reliable
  - Easy to implement
  - Doesn't require persistent connections
  - Works well when updates are infrequent
- 

## 12. Senior Engineer Insight (4–10 YOE)

A senior engineer doesn't ask:

▮ "Can we use Polling?"

They ask:

▮ "How frequently does data change?"

If data changes:

- once every few hours → Polling is perfectly fine.
- once every few minutes → Polling or Long Polling may work.
- every few milliseconds (chat, trading, collaboration) → Polling is the wrong tool.

**Choosing the simplest solution that satisfies the requirement is often better than introducing unnecessary complexity.** Not every problem needs WebSockets.